

Crime

pau, fer, y teo

Abstract

Introduction

Literature Review

Data and descriptive Statistics

importamos la bd

```
#install.packages("pder")  
library(pder)
```

Cargando paquete requerido: plm

```
data("Mafia")
```

Limpiamos la bd

```
# Cargar librerías  
library(sf)
```

Linking to GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1; sf_use_s2() is TRUE

```
library(dplyr)
```

Adjuntando el paquete: 'dplyr'

The following objects are masked from 'package:plm':

between, lag, lead

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
# Cargar shapefile de Italia
italy <- st_read("it_shp/it.shp", quiet = TRUE)

# Tabla de equivalencias entre nombres del shapefile y del dataset Mafia
equivalencias <- data.frame(
  shp_name = c("Turin", "Aoste", "Mantova", "Bozen", "Padova",
               "Reggio Emilia", "Forlì-Cesena", "Massa-Carrara",
               "Firenze", "Pesaro e Urbino", "Reggio Calabria",
               "Siracusa", "Oristrano"),
  mafia_name = c("Torino", "Aosta", "Mantua", "Bolzano", "Padua",
                 "Reggio Nell'Emilia", "Forli' - Cesena", "Massa Carrara",
                 "Florence", "Pesaro E Urbino", "Reggio Di Calabria",
                 "Syracuse", "Oristano")
)

# Filtrar y renombrar provincias para que coincidan con Mafia
italy_clean <- italy |>
  left_join(equivalencias, by = c("name" = "shp_name")) |>
  mutate(province = ifelse(is.na(mafia_name), name, mafia_name)) |>
  filter(province %in% unique(Mafia$province))
plot(st_geometry(italy_clean))
```



Descriptivas :)

```
library(knitr)
library(tidyr)

# Tabla de estadísticas descriptivas
Mafia |>
  select(y, g, cd, cds1, cds2, u1, u2, mafiosi, murder) |>
  summarise(across(everything(), list(
    Mean = ~mean(., na.rm = TRUE),
    SD   = ~sd(., na.rm = TRUE),
    Min  = ~min(., na.rm = TRUE),
    Max  = ~max(., na.rm = TRUE)
  ))) |>
  pivot_longer(everything(),
    names_to = c("Variable", ".value"),
    names_sep = "_") |>
  mutate(Description = c(
    "% growth of real per-capita value added",
    "Variation of per-capita public investment / lagged GDP",
    "Municipalities under external commissioners",
    "Same as cd (decree in first semester)",
    "Same as cd (avg days < 180)",
    "Change in log per-capita employment",
```

```

    "Change in log per-capita hours of unemployment insurance",
    "1st diff. of mafia-type association reports",
    "1st diff. of mafia-related murder reports"
  )) |>
  select(Variable, Description, Mean, SD, Min, Max) |>
  kable(digits = 3, caption = "Descriptive Statistics")

```

Variable	Description	Mean	SD	Min	Max
y	% growth of real per-capita value added	1.050	2.414	-9.721	9.449
g	Variation of per-capita public investment / lagged GDP	-0.067	1.277	-9.920	10.138
cd	Municipalities under external commissioners	0.005	0.051	0.000	1.209
cds1	Same as cd (decree in first semester)	0.002	0.023	0.000	0.576
cds2	Same as cd (avg days < 180)	0.003	0.040	0.000	1.209
u1	Change in log per-capita employment	-0.110	0.390	-2.137	1.469
u2	Change in log per-capita hours of unemployment insurance	-0.005	0.036	-0.169	0.106
mafiosi	1st diff. of mafia-type association reports	0.003	0.124	-1.162	1.253
murder	1st diff. of mafia-related murder reports	0.000	0.009	-0.155	0.144

Mapas

```

library(ggplot2)

# Promediar variables por provincia
mafia_avg <- Mafia |>
  group_by(province) |>
  summarise(
    y      = mean(y,      na.rm = TRUE),
    g      = mean(g,      na.rm = TRUE),
    mafiosi = mean(mafiosi, na.rm = TRUE),
    murder  = mean(murder, na.rm = TRUE),
    cds1    = sum(cds1,    na.rm = TRUE),
    cds2    = sum(cds2,    na.rm = TRUE)
  )

# Join con shapefile
italy_mafia <- italy_clean |>

```

```

left_join(mafia_avg, by = "province")

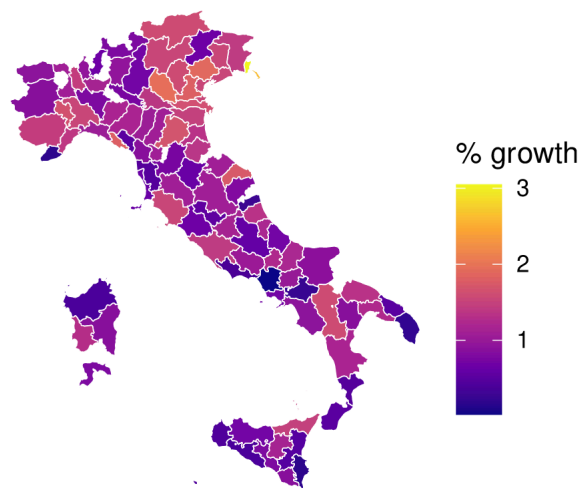
# mapa
mapa_italia <- function(variable, titulo, leyenda) {
  ggplot(italy_mafia) +
    geom_sf(aes(fill = {{ variable }}), color = "white", linewidth = 0.2) +
    scale_fill_viridis_c(option = "plasma", name = leyenda) +
    labs(title = titulo,
         caption = "Source: Mafia dataset (pder). Average 1986-1999.") +
    theme_void() +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold"),
      legend.position = "right"
    )
}

# Mapas individuales
mapa_y <- mapa_italia(y, "GDP per capita growth (y)", "% growth")
mapa_g <- mapa_italia(g, "Public Spending variation (g)",
"variation")
mapa_mafiosi <- mapa_italia(mafiosi, "Mafia activity (mafiosi)", "1st
diff")
mapa_murder <- mapa_italia(murder, "Mafia-related murders", "1st
diff")
mapa_cds1 <- mapa_italia(cds1, "Municipalities under commissioners (cds1)",
"count")
mapa_cds2 <- mapa_italia(cds2, "Municipalities under commissioners (cds2)",
"count")

# Ver todos
mapa_y; mapa_g; mapa_mafiosi; mapa_murder; mapa_cds1; mapa_cds2

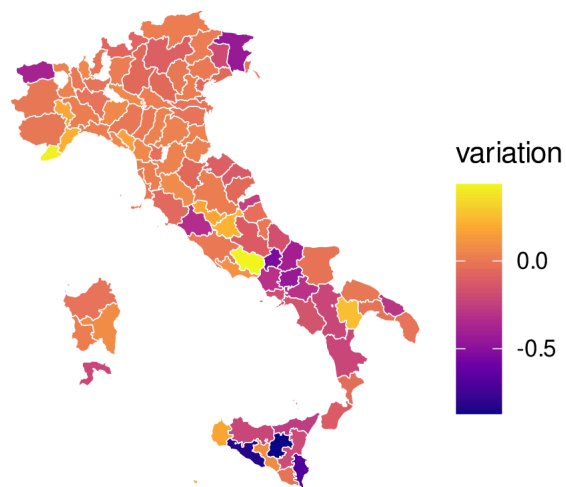
```

GDP per capita growth (y)



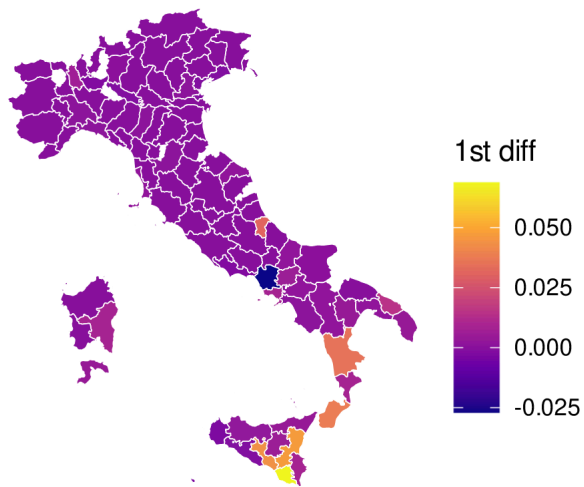
Source: Mafia dataset (pder). Average 1986-1999.

Public Spending variation (g)



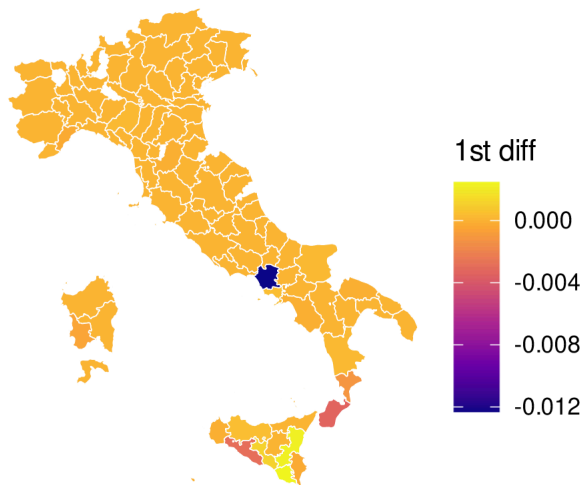
Source: Mafia dataset (pder). Average 1986-1999.

Mafia activity (mafiosi)



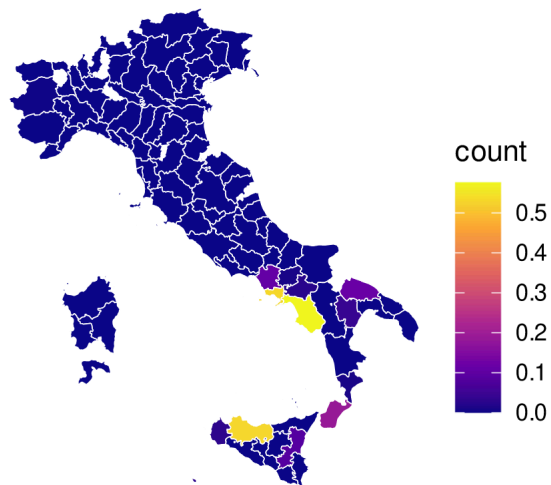
Source: Mafia dataset (pder). Average 1986-1999.

Mafia-related murders



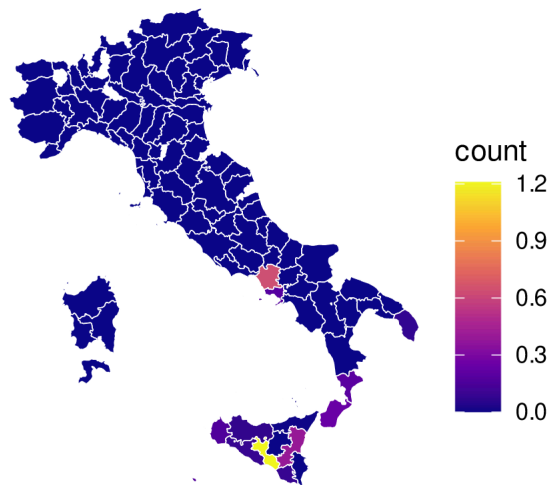
Source: Mafia dataset (pder). Average 1986-1999.

Municipalities under commissioners (cds1)



Source: Mafia dataset (pder). Average 1986-1999.

Municipalities under commissioners (cds2)



Source: Mafia dataset (pder). Average 1986-1999.

Matriz de pesos

```
library(spdep)
```

Cargando paquete requerido: spData


```
To access larger datasets in this package, install the spDataLarge
package with: `install.packages('spDataLarge',
repos='https://nowosad.github.io/drat/', type='source')`
```

```
library(sf)

# Centroides y matriz W por knn=4
coords <- st_centroid(italy_clean) |> st_coordinates()
```

Warning: st_centroid assumes attributes are constant over geometries

```
italy_nb <- knn2nb(knearneigh(coords, k = 4))
italy_W <- nb2listw(italy_nb, style = "W")
```

Test de Moran para autocorrelación espacial

```
# Moran's I test para y y g
moran_y <- moran.test(italy_mafia$y, italy_W)
moran_g <- moran.test(italy_mafia$g, italy_W)

moran_y
```

Moran I test under randomisation

```
data: italy_mafia$y
weights: italy_W
```

Moran I statistic standard deviate = 3.8462, p-value = 5.999e-05

alternative hypothesis: greater

sample estimates:

Moran I statistic	Expectation	Variance
0.246575985	-0.010638298	0.004472314

```
moran_g
```

Moran I test under randomisation

```
data: italy_mafia$g
weights: italy_W
```

```
Moran I statistic standard deviate = 3.84, p-value = 6.152e-05
alternative hypothesis: greater
sample estimates:
Moran I statistic      Expectation      Variance
      0.243289905      -0.010638298      0.004372868
```

Scatter plot

```
library(ggplot2)

moran_plot_gg <- function(var, W, titulo, color) {
  lag_var <- lag.listw(W, var)
  df <- data.frame(x = var, y = lag_var)

  # p-value del test de Moran
  mt <- moran.test(var, W)
  pval <- format(mt$p.value, scientific = TRUE, digits = 3)
  moran_i <- round(mt$estimate[1], 3)

  ggplot(df, aes(x = x, y = y)) +
    geom_point(color = color, alpha = 0.7) +
    geom_smooth(method = "lm", se = FALSE, color = "black", linewidth = 0.8) +
    geom_vline(xintercept = mean(var), linetype = "dashed", color = "gray50") +
    geom_hline(yintercept = mean(lag_var), linetype = "dashed", color = "gray50")
  +
    labs(title = titulo,
         x = "Variable", y = "Spatially lagged variable",
         subtitle = paste0("Moran's I = ", moran_i, " | p-value = ", pval)) +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.5, face = "bold"),
          plot.subtitle = element_text(hjust = 0.5, color = "gray40"))
}

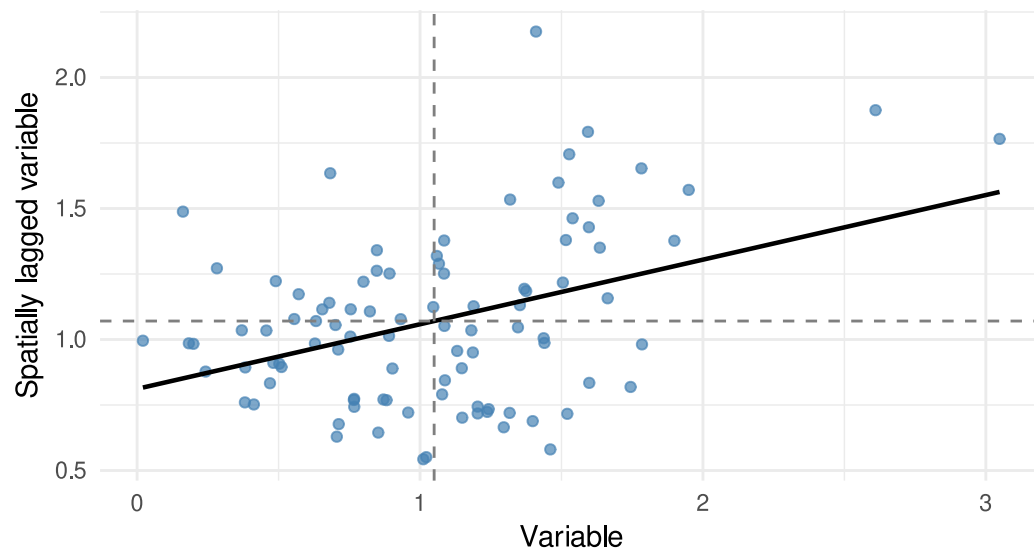
# Generar plots
mp_y <- moran_plot_gg(italy_mafia$y, italy_W,
                      "Moran Scatter Plot: GDP growth (y)", "steelblue")
mp_g <- moran_plot_gg(italy_mafia$g, italy_W,
                      "Moran Scatter Plot: Public Spending (g)", "coral")

mp_y
```

```
`geom_smooth()` using formula = 'y ~ x'
```

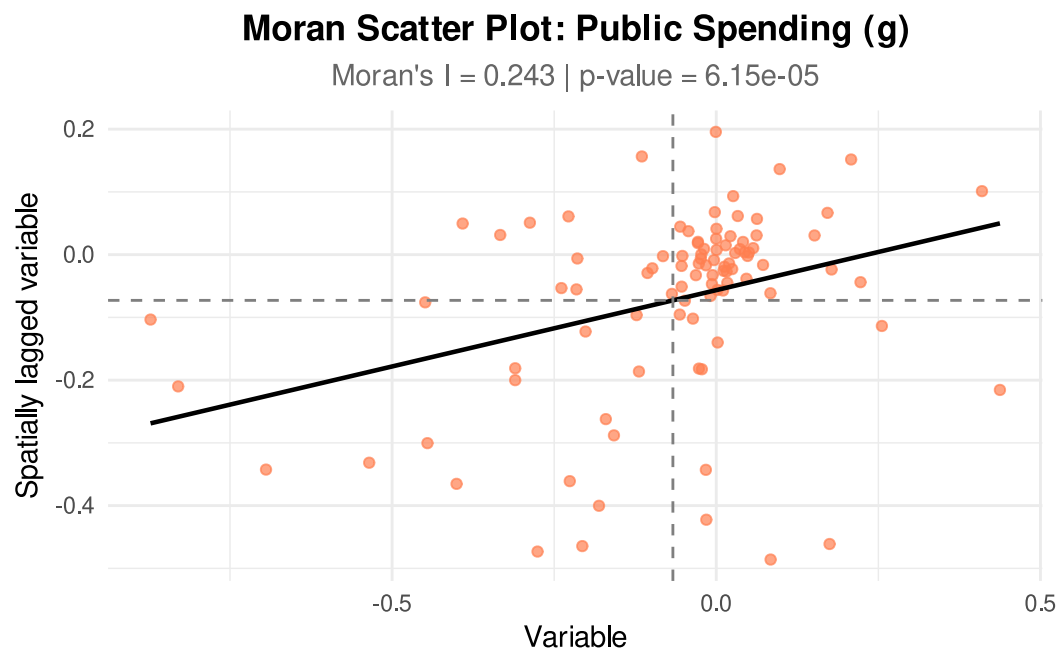
Moran Scatter Plot: GDP growth (y)

Moran's I = 0.247 | p-value = 6e-05



mp_g

```
`geom_smooth()` using formula = 'y ~ x'
```



Econometric Model

Result

Conclusions

References